Applied Machine Learning



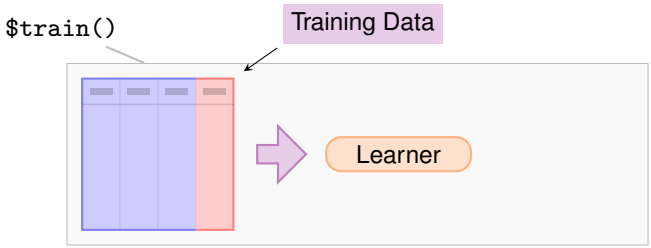
Machine Learning in R
MLR3 Learning & Evaluation



Learning goals

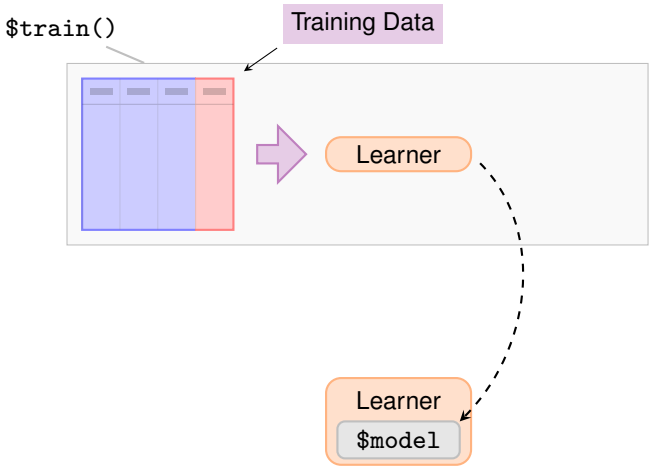
- Working with Learners in MLR3
- Train, Predict, Evaluate workflow
- Understanding model performance evaluation

LEARNING ALGORITHMS

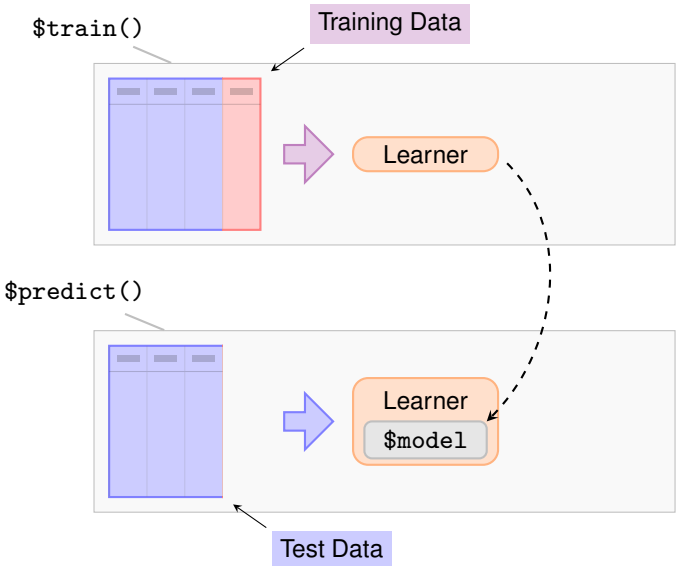




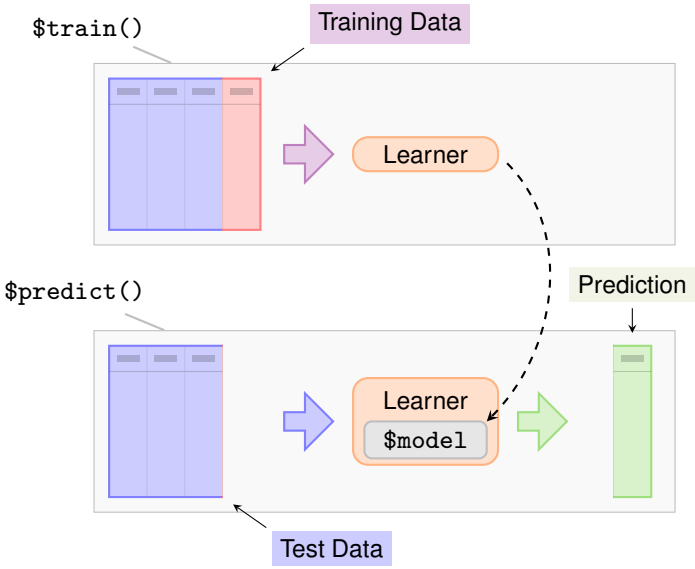
LEARNING ALGORITHMS



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LEARNING ALGORITHMS

- Get a `Learner` provided by `mlr`

```
learner = lrn("classif.rpart")
```



LEARNING ALGORITHMS



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learner = lrn("classif.rpart")
```

- Train the `Learner`

```
learner$train(task)
```

LEARNING ALGORITHMS



- Get a `Learner` provided by `mlr`

```
learner = lrn("classif.rpart")
```

- Train the `Learner`

```
learner$train(task)
```

- The `$model` is the `rpart` model: a decision tree

```
print(learner$model)

#> n= 150
#>
#> node), split, n, loss, yval, (yprob)
#>       * denotes terminal node
#>
#> 1) root 150 100 setosa (0.333 0.333 0.333)
#>   2) Petal.Length< 2.4 50   0 setosa (1.000 0.000 0.000) *
#>   3) Petal.Length>=2.4 100  50 versicolor (0.000 0.500 0.500)
#>     6) Petal.Width< 1.8 54   5 versicolor (0.000 0.907 0.093) *
#>     7) Petal.Width>=1.8 46   1 virginica (0.000 0.022 0.978) *
```

HYPERPARAMETERS



- Learners have *hyperparameters* that control their behavior

```
as.data.table(learner$param_set)[, 1:6]

#>                 id    class lower upper      levels nlevels
#>              <char>   <char> <num> <num>      <list>   <num>
#>  1:             cp ParamDbl     0     1                 Inf
#>  2:     keep_model ParamLgl    NA    NA TRUE,FALSE       2
#>  3:     maxcompete ParamInt     0   Inf                 Inf
#>  4:       maxdepth ParamInt     1    30                  30
#>  5:   maxsurrogate ParamInt     0   Inf                 Inf
#>  6:      minbucket ParamInt     1   Inf                 Inf
#>  7:       minsplit ParamInt     1   Inf                 Inf
#>  8: surrogatestyle ParamInt     0     1                   2
#>  9:   usesurrogate ParamInt     0     2                   3
#> 10:           xval ParamInt     0   Inf                 Inf
```

HYPERPARAMETERS



- Changing hyperparameters after the creation of a `Learner` object:

```
learner$param_set$values = list(maxdepth = 1, xval = 0)
```

- Changing hyperparameters when the `Learner` object is created:

```
learner = lrn("classif.rpart", maxdepth = 1, xval = 0)
```

HYPERPARAMETERS



- Changing hyperparameters after the creation of a `Learner` object:

```
learner$param_set$values = list(maxdepth = 1, xval = 0)
```

- Changing hyperparameters when the `Learner` object is created:

```
learner = lrn("classif.rpart", maxdepth = 1, xval = 0)
```

- The `Learner` behavior changes and gives a smaller decision tree:

```
learner$train(task)
print(learner$model)
#> n= 150
#>
#> node), split, n, loss, yval, (yprob)
#>       * denotes terminal node
#>
#> 1) root 150 100 setosa (0.33 0.33 0.33)
#>   2) Petal.Length< 2.4 50   0 setosa (1.00 0.00 0.00) *
#>   3) Petal.Length>=2.4 100  50 versicolor (0.00 0.50 0.50) *
```

PREDICTION



- Let's make a prediction for some new data, e.g.:

```
new_data
#   Sepal.Length Sepal.Width Petal.Length Petal.Width
# 1            4           3            2           1
# 2            2           2            3           2
```

PREDICTION



- Let's make a prediction for some new data, e.g.:

```
new_data
#   Sepal.Length Sepal.Width Petal.Length Petal.Width
# 1            4           3            2           1
# 2            2           2            3           2
```

- To do so, we call the `$predict_newdata()` method using the new data:

```
prediction = learner$predict_newdata(new_data)
```

PREDICTION



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- To do so, we call the `$predict_newdata()` method using the new data:

```
prediction = learner$predict_newdata(new_data)
```

- We get a `Prediction` object:

```
prediction
#> <PredictionClassif> for 2 observations:
#>  row_ids truth   response
#>        1  <NA>     setosa
#>        2  <NA> versicolor
```

PREDICTION



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```
new_data
#   Sepal.Length Sepal.Width Petal.Length Petal.Width
# 1            4           3            2           1
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```

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```
prediction
#> <PredictionClassif> for 2 observations:
#>  row_ids truth    response
#>        1  <NA>      setosa
#>        2  <NA> versicolor
```

PREDICTION



- We can make the `Learner` predict *probabilities* when we set `predict_type`:

```
learner$predict_type = "prob"
learner$predict_newdata(new_data)
# <PredictionClassif> for 2 observations:
#  row_ids truth  response prob.setosa prob.versicolor
#        1  <NA>    setosa           1             0.0
#        2  <NA> virginica           0             0.5
#  prob.virginica
#             0.0
#             0.5
```

PREDICTION



What exactly is a `Prediction` object?

- Contains predictions and offers useful access fields / methods

PREDICTION



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- Contains predictions and offers useful access fields / methods

⇒ Use `as.data.table()` to extract data

```
as.data.table(prediction)
#>    row_ids  truth   response
#>      <int> <fctr>     <fctr>
#> 1:       1   <NA>     setosa
#> 2:       2   <NA> versicolor
```

PREDICTION



What exactly is a `Prediction` object?

- Contains predictions and offers useful access fields / methods

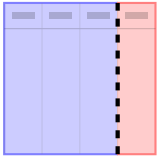
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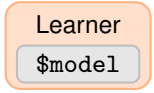
```
as.data.table(prediction)
#>    row_ids  truth   response
#>      <int> <fctr>     <fctr>
#> 1:       1   <NA>     setosa
#> 2:       2   <NA> versicolor
```

⇒ Active bindings and functions that give further information: `$response`, `$truth`, ...

```
prediction$response
#> [1] setosa     versicolor
#> Levels: setosa versicolor virginica
```

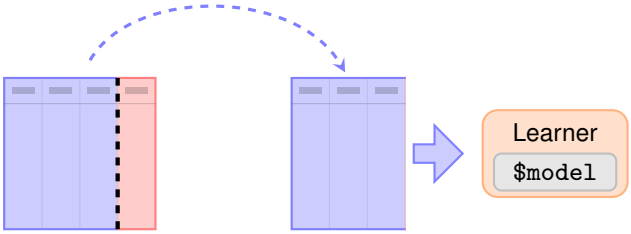
PERFORMANCE EVALUATION





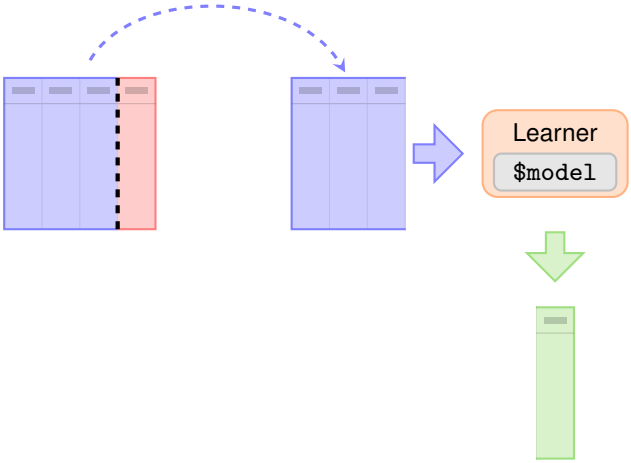


PERFORMANCE EVALUATION



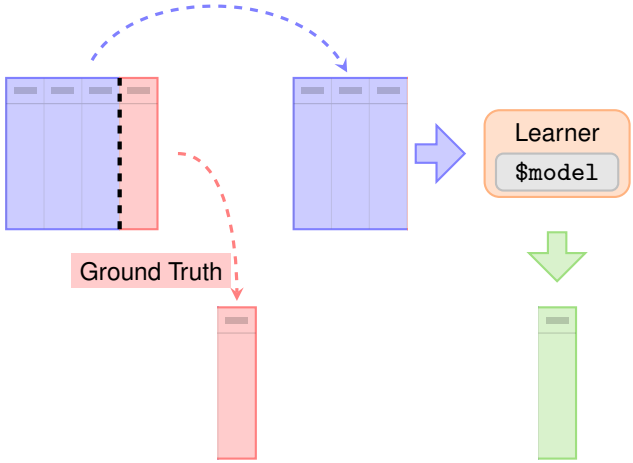


PERFORMANCE EVALUATION



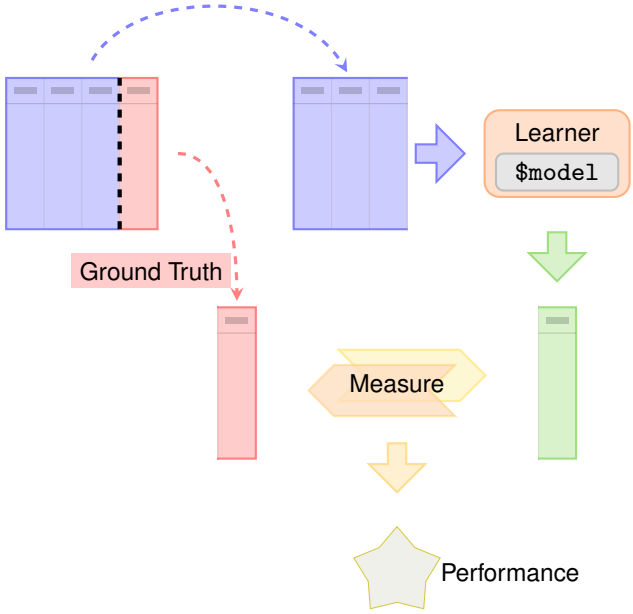


PERFORMANCE EVALUATION





PERFORMANCE EVALUATION



PERFORMANCE EVALUATION



- Prediction ‘Task‘ with known data

```
known_truth_task$data()
#     Species Petal.Length Petal.Width Sepal.Length Sepal.Width
#      <fctr>        <num>       <num>        <num>       <num>
# 1:  setosa            2           1            4           3
# 2:  setosa            3           2            2           2
```

PERFORMANCE EVALUATION



- Prediction ‘Task‘ with known data

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#    Species Petal.Length Petal.Width Sepal.Length Sepal.Width
#     <fctr>        <num>       <num>        <num>       <num>
# 1:  setosa            2           1            4           3
# 2:  setosa            3           2            2           2
```

- Predict again

```
pred = learner$predict(known_truth_task)
pred
#> <PredictionClassif> for 2 observations:
#>  row_ids  truth  response
#>        1 setosa    setosa
#>        2 setosa virginica
```

PERFORMANCE EVALUATION



- Prediction ‘Task‘ with known data

```
known_truth_task$data()
#     Species Petal.Length Petal.Width Sepal.Length Sepal.Width
#      <fctr>        <num>       <num>        <num>       <num>
# 1:  setosa            2           1            4           3
# 2:  setosa            3           2            2           2
```

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```

- Score the prediction

```
pred$score(msr("classif.ce"))
#> classif.ce
#>        0.5
```

PERFORMANCE EVALUATION



- Prediction ‘Task‘ with known data

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known_truth_task$data()
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